C2 Solutions: exponential and logarithmic functions

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Exact answers were checked with SymPy and approximations with Python.

Practice

1.

a. $2^6 = 64$, so ${}^2\log 64 = 6$.
b. $5^{-3} = \frac{1}{125}$, so -3.
c. $8^{2/3} = (\sqrt[3]{8})^2 = 4$, so $\frac{2}{3}$. (Or: $8^y = 4 \Rightarrow 2^{3y} = 2^2 \Rightarrow y = \frac{2}{3}$.)
d. $\left(\frac{1}{3}\right)^{-3} = 27$, so -3.
e. $e^2\sqrt{e} = e^{5/2}$, so $\frac{5}{2}$.
f. $0.0001 = 10^{-4}$, so -4.

2a. ${}^2\log 12 + {}^2\log 6 - {}^2\log 9 = {}^2\log \dfrac{12 \cdot 6}{9} = {}^2\log 8 = 3$.

2b. $\ln x^3 - \ln x^2 + \ln \frac{4}{x} = \ln \dfrac{x^3 \cdot 4}{x^2 \cdot x} = \ln 4$. The expression is constant (for $x > 0$).

3a. $81 = 3^4$, so $x + 1 = 4$, $x = 3$. **3b.** $3^{2x} = 3^{3(x-2)}$, so $2x = 3x - 6$, $x = 6$. **3c.** $x = {}^7\log 20 = \dfrac{\log 20}{\log 7} \approx 1.54$. **3d.** $2^x \cdot 5^x = 10^x = 10^{-2}$, so $x = -2$.

4a. Domain $2x - 1 > 0$. By definition $2x - 1 = 3^2 = 9$, so $x = 5$ (in the domain).

4b. Domain: $x > 0$ and $x > -2$, so $x > 0$. Combine: ${}^2\log(x(x+2)) = 3$, so $x^2 + 2x = 8$, $(x+4)(x-2) = 0$. $x = -4$ is outside the domain. **Answer:** $x = 2$.

5. $9^x = (3^x)^2$. Put $u = 3^x > 0$: $u^2 - 4u + 3 = 0$, $(u-1)(u-3) = 0$, $u = 1$ or $u = 3$. So $x = 0$ or $x = 1$.

6a. $g = \left(\frac{1}{2}\right)^{1/30} \approx 0.9772$.

6b. $\left(\frac{1}{2}\right)^{t/30} = 0.05 \Rightarrow \frac{t}{30} = {}^{0.5}\log 0.05 = \dfrac{\ln 0.05}{\ln 0.5} \Rightarrow t = 30 \cdot \dfrac{\ln 0.05}{\ln 0.5} \approx 129.7$. About **130 years**.

7a. $1.06^T = 2 \Rightarrow T = {}^{1.06}\log 2 = \dfrac{\ln 2}{\ln 1.06} \approx 11.90$ years.

7b. Rule of 70: $\frac{70}{6} \approx 11.67$ years. Close, but a little low, because $\ln(1.06) \approx 0.058$ is a bit smaller than 0.06.

8a. $3 \cdot 2^{x+1} > 0$, and it takes every positive value, so $f(x) > -12$. Asymptote $y = -12$, range $(-12, \infty)$.

8b. $3 \cdot 2^{x+1} = 12 \Rightarrow 2^{x+1} = 4 = 2^2 \Rightarrow x = 1$.

8c. $y = 3 \cdot 2^{x+1} - 12 \Rightarrow 2^{x+1} = \dfrac{y+12}{3} \Rightarrow x + 1 = {}^2\log \dfrac{y+12}{3} \Rightarrow x = {}^2\log \dfrac{y+12}{3} - 1$. So $f^{-1}(x) = {}^2\log \dfrac{x+12}{3} - 1$, domain $x > -12$.

9a. $3x = \ln 7$, so $x = \frac{1}{3}\ln 7$.

9b. Domain $x > 2$. $\ln(x(x-2)) = \ln 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. **Answer:** $x = 4$ ($x = -2$ is outside the domain).

10. $[\mathrm{H}^+] = 10^{-\mathrm{pH}}$. Ratio $\dfrac{10^{-2.5}}{10^{-6.0}} = 10^{3.5} \approx 3162$. About 3200 times as large.

11a. $0.6^x = 0.2 \Rightarrow x = {}^{0.6}\log 0.2 = \dfrac{\ln 0.2}{\ln 0.6} \approx 3.15$. Since $0.6 < 1$, 0.6^x is decreasing, so $0.6^x > 0.2$ for $x < \dfrac{\ln 0.2}{\ln 0.6}$, i.e. $x < 3.15$ approximately.

11b. Domain $x > -2$. Equation: ${}^{1/3}\log(x+2) = -1 \iff x + 2 = \left(\frac{1}{3}\right)^{-1} = 3 \iff x = 1$. ${}^{1/3}\log$ is decreasing, so the left side is smaller than -1 when $x + 2 > 3$. **Answer:** $x > 1$.

CE-level

12a. $9 - x^2 > 0 \iff -3 < x < 3$.

12b. ln is one-to-one: $9 - x^2 = 5$, $x^2 = 4$, $x = \pm 2$ (both in the domain).

12c. ln is increasing: $9 - x^2 \geq 8 \iff x^2 \leq 1 \iff -1 \leq x \leq 1$. (Then automatically $9 - x^2 > 0$.)

12d. For all x in the domain, $9 - x^2 \leq 9$, with equality only at $x = 0$. Since ln is increasing, $f(x) \leq \ln 9 = \ln 3^2 = 2\ln 3$, and $f(0) = 2\ln 3$. So the maximum is $2\ln 3$.

13a. Reflection in the line $x = 2$ sends the point (x, y) to $(4 - x, y)$. So the reflected graph of f is $y = f(4 - x) = \ln(4 - x) = g(x)$.

13b. $\ln x = \ln(4 - x) \iff x = 4 - x \iff x = 2$. Point $(2, \ln 2)$. (As expected, it lies on the mirror line.)

13c. Domain $0 < x < 4$. $\ln(x(4-x)) = \ln 3 \iff 4x - x^2 = 3 \iff x^2 - 4x + 3 = 0 \iff x = 1$ or $x = 3$. Both lie in the domain.

14a. $0.85^T = \frac{1}{2} \Rightarrow T = \dfrac{\ln 0.5}{\ln 0.85} \approx 4.27$ hours.

14b. $0.85^t = e^{t \ln 0.85}$, so $-k = \ln 0.85$, $k = -\ln 0.85 = \ln \frac{20}{17} \approx 0.1625$.

14c. $12 \cdot 0.85^t = 1 \Rightarrow 0.85^t = \frac{1}{12} \Rightarrow t = \dfrac{\ln(1/12)}{\ln 0.85} = \dfrac{\ln 12}{\ln(20/17)} \approx 15.29$ hours. Since C is decreasing, $C > 1$ for $0 \leq t < 15.29$. It works for about **15 hours and 17 minutes**.

15a. By change of base (any base h, say 10): ${}^a\log b \cdot {}^b\log c = \dfrac{\log b}{\log a} \cdot \dfrac{\log c}{\log b} = \dfrac{\log c}{\log a} = {}^a\log c$. (We used $\log a \neq 0$ and $\log b \neq 0$, which holds since $a, b \neq 1$.)

15b. Apply 15a repeatedly: the product collapses to ${}^2\log 16 = 4$.

16a. $e^x(e^x - 4) = 0$. $e^x > 0$, so $e^x = 4$, $x = \ln 4 = 2\ln 2$.

16b. Put $u = e^x > 0$: $u^2 - 4u - 5 = 0$, $(u - 5)(u + 1) = 0$, $u = 5$. So $x = \ln 5$.

16c. $e^x(e^x - 4) < 0 \iff e^x - 4 < 0$ (since $e^x > 0$) $\iff e^x < 4 \iff x < \ln 4$.

17a. Two machines: intensity $2I$. $L = 10\log\dfrac{2I}{I_0} = 10\log 2 + 10\log\dfrac{I}{I_0} = 10\log 2 + 80 \approx 83.0$ dB.

17b. n machines give $80 + 10\log n$ dB. $80 + 10\log n = 100 \Rightarrow \log n = 2 \Rightarrow n = 100$ machines.

18a. $M = \frac{2}{3}\log\frac{E}{E_0} \Rightarrow \log\frac{E}{E_0} = \frac{3}{2}M \Rightarrow E = E_0 \cdot 10^{1.5M}$. Replacing M by $M+1$ multiplies E by $10^{1.5}$.

18b. Difference 1.4: factor $10^{1.5\cdot 1.4} = 10^{2.1} \approx 126$.

19a. Slope of the line: $\dfrac{4-1.5}{5-0} = 0.5$. So $\log N = 1.5 + 0.5t$ and $N = 10^{1.5}\cdot(10^{0.5})^t$. So $b = 10^{1.5} = 10\sqrt{10} \approx 31.62$ and $g = \sqrt{10} \approx 3.16$.

19b. $(\sqrt{10})^T = 2 \Rightarrow 0.5T = \log 2 \Rightarrow T = 2\log 2 = \log 4 \approx 0.602$ hours ≈ 36 minutes.

Challenge

20. ${}^2\log 3 > 0$ because $3 > 1$. Suppose ${}^2\log 3 = \frac{p}{q}$ with positive integers p, q. Then $2^{p/q} = 3$, so (raising to the power q) $2^p = 3^q$. The left side is even (as $p \geq 1$) and the right side is odd. Contradiction. So ${}^2\log 3$ is irrational. □

21. $31^{11} < 32^{11} = 2^{55} < 2^{56} = (2^4)^{14} = 16^{14} < 17^{14}$. So 17^{14} is larger.

22. Domain $x > 0$. Take log of both sides: $\log x \cdot \log x = \log 100 + \log x$. Put $u = \log x$: $u^2 - u - 2 = 0$, $(u-2)(u+1) = 0$, $u = 2$ or $u = -1$. So $x = 100$ or $x = 0.1$. Check: $100^2 = 10\,000 = 100 \cdot 100$. $0.1^{-1} = 10 = 100 \cdot 0.1$. Both correct.

23a. $x = 2$: $4 = 4$. $x = 4$: $16 = 16$.

23b. Let $h(x) = 2^x - x^2$. $h(-1) = \frac{1}{2} - 1 < 0$ and $h(0) = 1 > 0$. Both terms are continuous, so h has a zero between -1 and 0. With a calculator: $x \approx -0.77$ (more precisely -0.7667).

23c. *Negative x:* for $x < 0$, 2^x is increasing and x^2 is decreasing, so h is strictly increasing and has at most one zero there. $x = 0$: $h(0) = 1 \neq 0$. *Positive x:* for $x > 0$ the equation is equivalent (take ln) to $x\ln 2 = 2\ln x$, i.e. $k(x) = x\ln 2 - 2\ln x = 0$. A full proof uses calculus from stage D: $k'(x) = \ln 2 - \frac{2}{x}$, which is negative for $x < \frac{2}{\ln 2}$ and positive after it. So k first decreases and then increases, and has at most two zeros. We have found two ($x = 2$ and $x = 4$), so there are no more. In total there are exactly three solutions.

24. Domain $x > 0$, $x \neq 1$. Put $u = {}^2\log x$ (so $u \neq 0$). Then ${}^x\log 2 = \dfrac{1}{{}^2\log x} = \dfrac{1}{u}$. The equation becomes $u + \frac{1}{u} = \frac{5}{2}$, i.e. $2u^2 - 5u + 2 = 0$, $(2u-1)(u-2) = 0$, so $u = 2$ or $u = \frac{1}{2}$. **Answer:** $x = 4$ or $x = \sqrt{2}$.

25. $\left(\frac{1}{2}\right)^{t/5730} = 0.30 \Rightarrow t = 5730 \cdot \dfrac{\ln 0.3}{\ln 0.5} \approx 9953$. About **9950 years**.